

NONCLINICAL REPORT

Nonclinical Report No. VRC02

Efficacy of mRNA-1273 and Enhanced Disease in Aged BALB/c mice

SPONSOR:
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National Institute of Allergy and Infectious Diseases
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LIST OF ABBREVIATIONS

Abbreviation	Definition
ACE-2	angiotensin-converting enzyme 2
ANOVA	analysis of variance
CoV	coronavirus
COVID-19	coronavirus disease 2019
DSPC	1,2-distearoyl-sn-glycero-3-phosphocholine
ELISA	enzyme-linked immunosorbent assay
ERD	enhanced respiratory disease
GMP	Good Manufacturing Practice
HEK	human embryonic kidney
IC ₅₀	half maximal inhibitory concentration
IgG	immunoglobulin G
LNP	lipid nanoparticle
LOD	limit of detection
mRNA	messenger RNA
NIAID	National Institute of Allergy and Infectious Diseases
NIH	National Institutes of Health
PBS	phosphate-buffered saline
PEG	polyethylene glycol
PEG2000-DMG	1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol
PFU	plaque-forming units
RBD	receptor binding domain
RT-PCR	reverse transcription polymerase chain reaction
S	spike
S-2P	spike protein with 2 proline substitutions within the heptad repeat 1 domain
SARS-CoV-2	2019 novel coronavirus
SARS-CoV-2 MA10	mouse-adapted SARS-CoV-2 that has been optimized for increased replication and disease manifestation in mice by 10 in vivo passages
SARS-CoV DIV	double-inactivated severe acute respiratory syndrome coronavirus
SM-102	heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate
Th1	T-helper 1
Th2	T-helper 2
(b) (4)	(b) (4)
UTR	untranslated region
UV	ultraviolet
VRC	Vaccine Research Center

QUALITY ASSURANCE STATEMENT

The study was not conducted according to Good Laboratory Practice regulations.

1. SUMMARY

Nonclinical studies have demonstrated that coronavirus (CoV) spike (S) proteins are immunogenic and that 2019 novel coronavirus (SARS-CoV-2) prefusion-stabilized S protein with 2 proline substitutions within the heptad repeat 1 domain (S-2P)-based vaccines, including those based on messenger RNA (mRNA) delivery platforms, are protective in animals. In addition, previous clinical studies of vaccines targeting related CoVs and other viruses have demonstrated that mRNA-based vaccines are safe and immunogenic.

Despite prior clinical safety of other mRNA vaccines, vaccines against viral respiratory diseases should be evaluated to address the theoretical concern of enhanced respiratory disease (ERD). The potential for ERD is predicted for vaccines where immune responses are characterized by low neutralizing antibody responses and/or a predominant T helper 2 (Th2) mediated T-cell response and was prompted by preclinical observations in severe acute respiratory syndrome (SARS)-CoV ([Bolles et al 2011](#)) and Middle East respiratory syndrome-CoV vaccines ([Houser et al 2017](#)); feline infectious peritonitis virus and vaccines in cats ([Pedersen et al 2009](#); [Pedersen et al 2012](#); [Takano et al 2008](#)); as well as the enhanced disease seen with respiratory syncytial virus (RSV) ([Kim et al 1969](#)), measles ([Polack 2007](#)), and dengue vaccines in humans ([Smatti et al 2018](#)).

The objectives of Study VRC02 were to evaluate the immunogenicity of mRNA-1273 and protection from SARS-CoV-2 challenge and to determine if a subprotective level of immunity leads to increased viral loads or disease, compared with double-inactivated severe acute respiratory syndrome coronavirus (SARS-CoV DIV) in aged mice. Although vaccine studies in young wild-type mouse models are common, studies in aged mice are typically not performed. However, because the incidence and severity of coronavirus disease 2019 (COVID-19) is greater in the elderly population, viral challenge studies were performed in aged mice to develop the model. After challenge with a mouse-adapted SARS-CoV-2 strain, more severe disease symptoms, including greater weight loss and lung hemorrhage, were observed in aged mice than in young mice. Therefore, this aged mouse model was used to further evaluate mRNA-1273.

ModernaTX, Inc. is developing mRNA-1273, a novel lipid nanoparticle (LNP)-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 contains a single mRNA that encodes for the SARS-CoV-2 S glycoprotein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P). The mRNA-1273 Drug Product consists of the mRNA formulated in a mixture of 4 ionizable, structural, helper, and polyethylene glycol (PEG) lipids common to the Sponsor's mRNA vaccine platform: heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate (SM-102); cholesterol; 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC); and 1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol (PEG2000-DMG), respectively.

The test articles in this study consisted of mRNA-1273 Drug Product (WH2020_NatSP_2P, Tech dev DP) and SARS-CoV DIV (positive vaccine control predicted to drive responses associated with ERD). Phosphate-buffered saline (PBS) was used as a negative control.

Aged female BALB/c mice approximately 1-year-old were vaccinated with a suboptimal (0.1 µg) or optimal (1 µg) dose of mRNA-1273, 0.1 µg of SARS-CoV DIV, or PBS via intramuscular (IM) injection into the hind leg on a prime (Week 0)/boost (Week 3) schedule. At Week 2 (2 weeks post-prime) and Week 5 (2 weeks post-boost), sera were collected to measure binding immunoglobulin (Ig) G antibody titers against SARS-CoV-2 S-2P by enzyme-linked immunosorbent assay (ELISA) and neutralization activity against a SARS-CoV-2 pseudotyped lentivirus. Additionally, IgG2a to IgG1 antibody subclass ratios were calculated at Week 5.

At Week 7 (4 weeks post-boost), mice were challenged intranasally with 10^4 plaque-forming units (PFU) of SARS-CoV-2 MA10, an adapted SARS-CoV-2 virus strain that recapitulates the age-related disease severity observed in humans ([Leist et al 2020](#)). Body weight was measured through Day 4 post-challenge. Lung and nasal turbinate samples were collected on Day 2 and Day 4 post-challenge to evaluate gross hemorrhage in the lung lobes, viral titers by plaque assay, lung cytokine RNA levels by reverse transcription polymerase chain reaction (RT-PCR), and histology by hematoxylin and eosin staining.

mRNA-1273 is highly immunogenic in aged mice. mRNA-1273 elicited robust binding and neutralizing antibody responses in a dose-dependent manner after a prime/boost immunization schedule. Post-boost IgG antibody titers and neutralizing antibody responses were significantly higher in mice immunized with 0.1 and 1 µg of mRNA-1273 than in mice in the SARS-CoV DIV control group.

In mice, a ratio of high IgG1 antibody subclass with low or no IgG2a antibody subclass is associated with a T-helper 2 (Th2)-directed response, whereas a balanced ratio of IgG2a and IgG1 antibody subclasses is associated with a T-helper 1 (Th1)-directed response. Post-boost, immunization with 0.1 and 1 µg of mRNA-1273 elicited both IgG2a and IgG1 subclass S-2P-binding antibodies with a balanced IgG2a/IgG1 antibody subclass ratio. Notably, the IgG2a and IgG1 titers were higher in the 0.1 and 1 µg mRNA-1273 groups than in the SARS-CoV DIV control group. The IgG2a antibody subclass titers were not detectable in the SARS-CoV DIV group; therefore, the IgG2a/IgG1 ratio could not be calculated. These results indicate that unlike SARS-CoV DIV, which induces a Th2-directed response that has been associated with ERD ([Bolles et al 2011](#)), mRNA-1273 is associated with a Th1-directed response that should not promote ERD.

Weight loss was not observed in mice vaccinated with 1 µg of mRNA-1273 through Day 4 post-challenge. Weight loss was observed in mice vaccinated with PBS, SARS-CoV DIV, and 0.1 µg of mRNA-1273; however, the weight loss observed in the 0.1 µg mRNA-1273 group was

less than that observed in the SARS-CoV DIV and PBS groups, indicating that a suboptimal dose (0.1 µg) of mRNA-1273 did not enhance weight loss.

mRNA-1273 protected aged BALB/c mice from SARS-CoV-2 infection in a dose-dependent manner. Immunization with 1 µg of mRNA-1273 completely protected mice from viral replication on Day 2 and Day 4 post-challenge, with substantially lower viral titers in the lungs and nasal turbinates compared with all other groups. In fact, viral titers in the lungs and nasal turbinates were below the limit of detection (LOD) at Day 4 post-challenge. Immunization with 0.1 µg of mRNA-1273 provided partial protection, as viral replication was observed in the lungs and nasal turbinates, but the mean viral titers were lower than those in the SARS-CoV DIV and PBS groups on Day 2 and Day 4 post-challenge.

No lung hemorrhage was observed in aged mice immunized with 1 µg of mRNA-1273 at Day 2 or Day 4 post-challenge. Lung hemorrhage was observed in mice in the 0.1 µg mRNA-1273, SARS-CoV DIV, and PBS groups on Day 4 post-challenge; however, the mean lung hemorrhage scores were lower in mice immunized with 0.1 µg of mRNA-1273 than in mice in the SARS-CoV DIV and PBS groups, indicating that a suboptimal dose (0.1 µg) of mRNA-1273 did not lead to lung hemorrhage enhancement.

The RT-PCR analysis showed that SARS-CoV DIV induced increased levels of lung cytokines and chemokines (Th2-associated cytokines and chemokines, Th1-associated cytokines, and inflammatory cytokines), and the expression levels of lung cytokines and chemokines were substantially higher at Day 4 post-challenge in the SARS-CoV group than the levels in mice immunized with mRNA-1273. The expression levels of these cytokines in the SARS-CoV DIV group were generally higher than in the PBS group, whereas the expression levels in the 0.1 and 1 µg mRNA-1273 groups were generally lower than those in the PBS group, indicating that SARS-CoV DIV stimulates an enhanced Th2-associated cytokine response with the potential for respiratory disease enhancement, whereas mRNA-1273 induces a Th1-directed noninflammatory immune response suggestive of reduced respiratory disease.

Lung histopathology and immunochemistry showed no alterations related to ERD or inflammation and no virus antigen-positive cells in mice immunized with 1 µg of mRNA-1273 on Day 2 or Day 4 post-challenge. Lung sections from mice immunized 0.1 µg of mRNA-1273 showed inflammation and SARS-CoV-2-antigen-positive cells on Day 2 post-challenge; however, virus antigen-positive cells were substantially reduced on Day 4 post-challenge. The SARS-CoV DIV and PBS groups showed marked alterations related to enhanced disease and inflammation with abundant SARS-CoV-2 virus antigen-positive cells.

Overall, mRNA-1273 is immunogenic and efficacious and does not promote ERD in aged mice.

2. DESCRIPTION OF TEST PROCEDURES

2.1. Conduct of Study

2.1.1. Sponsor

Vaccine Research Center (VRC)
National Institute of Allergy and Infectious Diseases (NIAID)
National Institutes of Health (NIH)
Bethesda, MD 20892

2.1.2. Performing Laboratory

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(b) (4)



2.1.3. Study Number

Study number (b) (4) -Study 2 (VRC02).

2.1.4. Purpose of the Study

The objectives of this study were to evaluate the immunogenicity of mRNA-1273 and protection from SARS-CoV-2 challenge and to determine if a subprotective level of immunity leads to increased viral loads or disease, compared with SARS-CoV DIV in an aged mouse model.

2.1.5. Institutional Animal Care and Use Committee Approval

All animal experiments described in this study report were approved by the Animal Care and Use Committee of the VRC (Protocol (b) (4) Study 2) and conducted in compliance with US NIH regulations.

2.1.6. Study Director

(b) (6)

Viral Pathogenesis Laboratory, VRC, NIH
Building 40, Room 2608
Bethesda, MD 20892

2.1.7. Key Personnel

Role	Study VRC02
Study coordinator	(b) (6) Viral Pathogenesis Laboratory, VRC, NIH Building 40, Room 2608 Bethesda, MD 20892
Analytical assays: ELISA	(b) (6) Viral Pathogenesis Laboratory, VRC, NIH Building 40, Room 2608 Bethesda, MD 20892
Analytical assays: Pseudovirus neutralization assay	(b) (6) Viral Pathogenesis Laboratory, VRC, NIH Building 40, Room 2608 Bethesda, MD 20892
Analytical assays: Cytokine analysis and viral load	(b) (4), (b) (6)
Analytical assays: Pathology	(b) (6) Infectious Disease Pathogenesis Section (IDPS), NIH/NIAID/ Contract Management Branch 12735 Twinbrook Parkway, Room 2W-01 Rockville, MD 20852-1770
Report compilation	PPD 3900 Paramount Parkway Morrisville, NC 27560

2.1.8. Dates of Study Performance

Schedule of Events	Study VRC02
Test article administration:	
Prime dose	08 May 2020 (Week 0)
Boost dose	29 May 2020 (Week 3)
Serum collection	22 May 2020 (Week 2) 12 Jun 2020 (Week 5)
Challenge with SARS-CoV-2	26 Jun 2020 (Week 7)
Lung and nasal turbinates sectioned	Day 2 (28 Jun 2020) and Day 4 (30 Jun 2020) after SARS-CoV-2 challenge

2.1.9. Records Maintained

Data are archived in the Viral Pathogenesis Laboratory (VRC, NIH) database, which is password protected and can be accessed only by authorized personnel. The server is maintained by NIH information technology staff and is backed up remotely on a daily basis.

2.2. Test Articles

The test articles in this study consisted of 0.1 and 1 µg of mRNA-1273 Drug Product and 0.1 µg of SARS-CoV DIV (positive vaccine control predicted to drive responses associated with ERD). Phosphate-buffered saline was used as a negative control.

2.2.1. mRNA-1273

mRNA-1273 contains a single mRNA that encodes SARS-CoV-2 S-2P formulated into a mixture of 4 lipids: SM-102, cholesterol, DSPC, and PEG2000-DMG. A summary of the dose formulation analysis results is presented in [Table 1](#).

The mRNA-1273 Drug Product used in this study was an mRNA formulation (WH2020_NatSP_2P, Tech dev DP, AMPDP-200005) prepared with the same method as the Good Manufacturing Practice (GMP) mRNA-1273 clinical Drug Product ([Appendix 1](#)). This product is a novel LNP-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 encodes for the full-length S protein of SARS-CoV-2, modified to introduce 2 proline residues to stabilize the S protein into a prefusion conformation (S-2P). mRNA-1273 consists of mRNA encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process described previously ([Hassett et al 2019](#)).

Preclinical mRNA-1273 vaccine was manufactured by ModernaTX, Inc. A sequence-optimized mRNA encoding prefusion-stabilized SARS-CoV-2 S-2P protein was synthesized in vitro using

an optimized T7 RNA polymerase-mediated transcription reaction with complete replacement of uridine by N¹m-pseudouridine (Nelson et al 2020). The reaction included a DNA template containing the immunogen open-reading frame flanked by 5' untranslated region (UTR) and 3' UTR sequences and was terminated by an encoded polyA tail. After transcription, the Cap 1 structure was added to the 5' end using vaccinia capping enzyme (New England Biolabs) and vaccinia 2'O methyltransferase (New England Biolabs). The mRNA was purified (b) (4)

(b) (4) buffer exchanged (b) (4)

sterile filtered, and kept frozen at –20°C until further use.

mRNA-1273 was encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process as previously described (Hassett et al 2019). (b) (4)

(b) (4)

The mixture was neutralized with tris(hydroxymethyl)aminomethane hydrochloride (Tris HCl), pH 7.5, sucrose was added as a cryoprotectant, and the final solution was sterile filtered. Vials were filled with mRNA-1273 formulated LNP and stored frozen at –70°C until further use. The Drug Product underwent analytical characterization, which included the determination of particle size and polydispersity, encapsulation, mRNA purity, double-stranded RNA content, osmolality, pH, endotoxin, and bioburden; the material was deemed acceptable for the in vivo study (Corbett et al 2020).

Table 1: Test Materials and Summary of Dose Formulation Analysis Results

Test Material	mRNA Description	LNP Lot No.	mRNA Lot No.	Size (nm)	PDI	EE%
mRNA-1273 Drug Product	mRNA SARS-CoV-2 spike protein ^a	(b) (4)	(b) (4)	(b) (4)	(b) (4)	(b) (4)

Abbreviations: EE% = encapsulation efficiency; LNP = lipid nanoparticle; mRNA = messenger RNA; PDI = polydispersity index; SARS-CoV-2 = 2019 novel coronavirus.

Note: All mRNA test material stocks were formulated in SM-102-containing lipid nanoparticles with a final storage buffer of 10.7 mM sodium acetate, 8.7% sucrose, and 20 mM tris(hydroxymethyl)aminomethane hydrochloride at pH 7.5 and stored frozen at –70°C.

The EE% is the percentage of mRNA (b) (4)

(b) (4)

(b) (4)

^a Full-length SARS-CoV-2 spike protein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P) and including a transmembrane domain for insertion into the cell membrane after translation.

^b Manufactured by ModernaTX, Inc. (01 Feb 2020).

2.2.2. SARS-CoV DIV

Animals were vaccinated with purified SARS-CoV that was inactivated with binary ethylenimine or formalin and UV (10 µg/mL formulated in alum) (NR-3883).

2.2.3. Phosphate-Buffered Saline

Phosphate-buffered saline (b) (4)) was purchased commercially.

2.3. Test System

Female BALB/c mice, approximately 1 year old, were obtained from Envigo (Indianapolis, IN, USA). Although vaccine studies in young wild-type mouse models are common, studies in aged mice are typically not performed. However, because the incidence and severity of COVID-19 is greater in the elderly population, viral challenge studies were performed in aged mice to develop the model. After challenge with a mouse-adapted SARS-CoV-2 strain, more severe disease symptoms, including greater weight loss and lung hemorrhage, were observed in aged mice than in young mice. Therefore, this aged mouse model was used to further evaluate mRNA-1273.

2.4. Methods

2.4.1. Study Design and Treatment Groups

Mice were treated according to the study design presented in [Table 2](#). Aged female BALB/c mice, approximately 1-year-old (n = 10/group), were vaccinated with a suboptimal (0.1 µg) or optimal (1 µg) dose of mRNA-1273, 0.1 µg of SARS-CoV DIV, or PBS via IM injection into the hind leg on a prime (Week 0)/boost (Week 3) schedule. At Week 2 (2 weeks post-prime) and Week 5 (2 weeks post-boost), sera were collected to measure binding IgG antibody titers against SARS-CoV-2 S-2P by ELISA and neutralization activity against a SARS-CoV-2 pseudotyped lentivirus. Additionally, IgG2a to IgG1 antibody subclass ratios were calculated at Week 5.

At Week 7 (4 weeks post-boost), all mice were challenged intranasally with 10³ PFU of SARS-CoV-2 MA10, an adapted SARS-CoV-2 virus strain that recapitulates the age-related disease severity observed in humans ([Leist et al 2020](#)). The SARS-CoV-2 MA10 virus has been adapted in mice by 10 in vivo passages in addition to the 2 targeted amino acid mutations (Q498T/P499Y) in the receptor binding domain that allow for mouse angiotensin-converting enzyme 2 (ACE-2) receptor binding ([Leist et al 2020](#)). These virus adaptations allow for an increased replication and disease manifestation in the upper and lower respiratory tract. Body weight was measured through Day 4 post-challenge. Lung and nasal turbinate samples were collected on Day 2 and Day 4 post-challenge to evaluate gross hemorrhage in the lung lobes, viral titers by plaque assay, lung cytokine RNA levels by RT-PCR, and histology by hematoxylin and eosin staining.

Table 2: Treatment Regimen for Study VRC02

Group	Treatment	Dose (µg)	ROA (Volume)	Schedule			Sample Collection	Experimental Assessments
				Prime	Boost	Challenge		
1	PBS	–	IM (50 µL)	Week 0	Week 3	Week 7	Sera (Week 2, Week 5)	IgG binding titers by ELISA and neutralization activity by pseudovirus neutralization assay
2	mRNA-1273	0.1						Weight measurement through Day 4 post-challenge
3	mRNA-1273	1						Lung hemorrhage, viral load in the lung and nasal turbinates by plaque assay, relative cytokine RNA levels by RT-PCR, lung pathology
4	SARS-CoV DIV	0.1					Lung and nasal turbinate samples (Day 2 and Day 4 post-challenge)	

Abbreviations: ELISA = enzyme-linked immunosorbent assay; IgG = immunoglobulin G; IM = intramuscular; PBS = phosphate-buffered saline; ROA = route of administration; RT-PCR = real-time reverse transcription polymerase chain reaction; SARS-CoV DIV = double-inactivated severe acute respiratory syndrome coronavirus.

2.4.2. SARS-CoV-2 S-2P Enzyme-Linked Immunosorbent Assay

(b) (4) 96-well ELISA plates (b) (4) were coated with 100 ng/well SARS-CoV-2 S-2P protein in 1× PBS at 4°C for 16 hours. To eliminate fold-on-specific binding from SARS-CoV-2 S-2P protein-immune mouse serum, 50 µg/mL of fold-on protein was added for 1 hour at room temperature. After standard washes and blocks, plates were incubated with serial dilutions of heat-inactivated sera for 1 hour at room temperature. Following washes, anti-mouse IgG, IgG1, or IgG2a–horseradish peroxidase conjugates (b) (4) were used as secondary antibodies, and 3,5,3',5'-tetramethylbenzidine (b) (4) was used as the substrate to detect antibody responses. Endpoint titer was calculated as the dilution that emitted an optical density exceeding 4× background (secondary antibody alone).

2.4.3. Lentivirus-Based Pseudovirus Neutralization Assay

(b) (4)

(b) (4) and then pseudovirus neutralization assays were performed (Pallesen et al 2017). Briefly,

heat-inactivated serum was mixed with pseudoviruses, incubated, and then added to AEC-2 receptor-expressing HEK293T cells obtained from (b) (6) at Scripps Research Institute. Cells were lysed after 72 hours of incubation, and luciferase activity (relative light units) was measured. Percent neutralization was normalized considering uninfected cells as 100% neutralization and cells infected with only pseudovirus as 0% neutralization. The half maximal inhibitory concentration (IC₅₀) titers were determined using a log (agonist) vs normalized response (variable slope) nonlinear function in Prism v8 (GraphPad).

2.4.4. Weight Measurement

Body weight was measured from the day of the SARS-CoV-2 MA10 challenge through Day 4 post-challenge (n = 10/group).

2.4.5. Lung Hemorrhage

Gross hemorrhage in the lungs was evaluated immediately after euthanasia. A score of 0 indicated no hemorrhage in any lobe, and a score of 4 indicated severe hemorrhage in all lung lobes.

2.4.6. Viral Load Assessment

Viral titers in lungs and nasal turbinates were assessed as previously described ([Dinnon et al 2020](#)).

2.4.6.1. Lung Cytokines and Chemokine Analysis

Lung cytokines were analyzed by a (b) (4) method.

2.4.7. Histology

Lung tissue samples were placed in 10% neutral buffered formalin until adequately fixed. Thereafter, tissues were trimmed to a thickness of 3 to 5 mm, processed, and paraffin embedded. Paraffin tissue blocks were subsequently sectioned (5 µm) and stained with hematoxylin and eosin. Sections were examined by a board-certified veterinary pathologist using an (b) (4) light microscope. An (b) (4) camera was used to take photomicrographs. Histopathology severity was evaluated using the histopathology scoring system presented in [Table 3](#).

Table 3: Histopathology Severity Grading System

Overall Severity grading	Severity of Neutrophil Infiltrate	Proportion of Airspace Area Affected
inflammation was absent (within normal histologic limits)	-	None
minimal-mild inflammation	-/+	≤10%
mild to moderate inflammation	+	≥10 – <25%
moderate to severe inflammation	++	≥25 – <50%
severe inflammation	+++	≥50%

Note: Histopathology severity grading was based on the severity scoring of neutrophil infiltrate and the proportion of airspace areas area affected. The level of severity was judged from – to +++, which represented none to severe. The proportion of airspace area affected was the ratio of the airspace area affected and the total airspace area of one view.

2.4.8. Immunohistochemistry

Immunohistochemical staining was performed using the formalin-fixed paraffin-embedded tissue sections (5 µm) obtained from lung tissues. Immunohistochemical staining used an anti-SARS-CoV-2 nucleocapsid rabbit polyclonal antibody (b) (4) at a dilution of 1:2000 or an anti-eosinophil major basic protein goat polyclonal antibody (b) (4) (b) (4) at a dilution of 1:500. Staining was carried out on the (b) (4) (b) (4) platform according to manufacturer-supplied protocols. Briefly, 5 µm-thick sections were deparaffinized and rehydrated. Heat-induced epitope retrieval was performed using (b) (4) (b) (4), pH 6.0, heated to 100°C for 20 minutes. Specimens were then incubated with hydrogen peroxide to quench endogenous peroxidase activity prior to applying the primary antibody. Detection with 3,3'-diaminobenzidine chromogen was completed using a (b) (4). Slides were finally cleared through gradient alcohol and xylene washes prior to mounting and placing cover slips.

Sections were examined by a board-certified veterinary pathologist using an (b) (4) light microscope. An (b) (4) camera was used to take photomicrographs.

2.5. Data Presentation and Analysis

Geometric means or means are represented by the heights of bars or symbols, and error bars represent the corresponding SD. Dotted lines indicate assay LODs. For IgG antibody responses and neutralizing activity, 2-way analysis of variance (ANOVA) with Sidak's or Tukey's multiple comparisons test was used to compare temporal and dose-dependent antibody responses post-boost, respectively. For viral load assessment, lung hemorrhage and, lung cytokines, groups were compared by 1-way ANOVA with Kruskal-Wallis post-test.

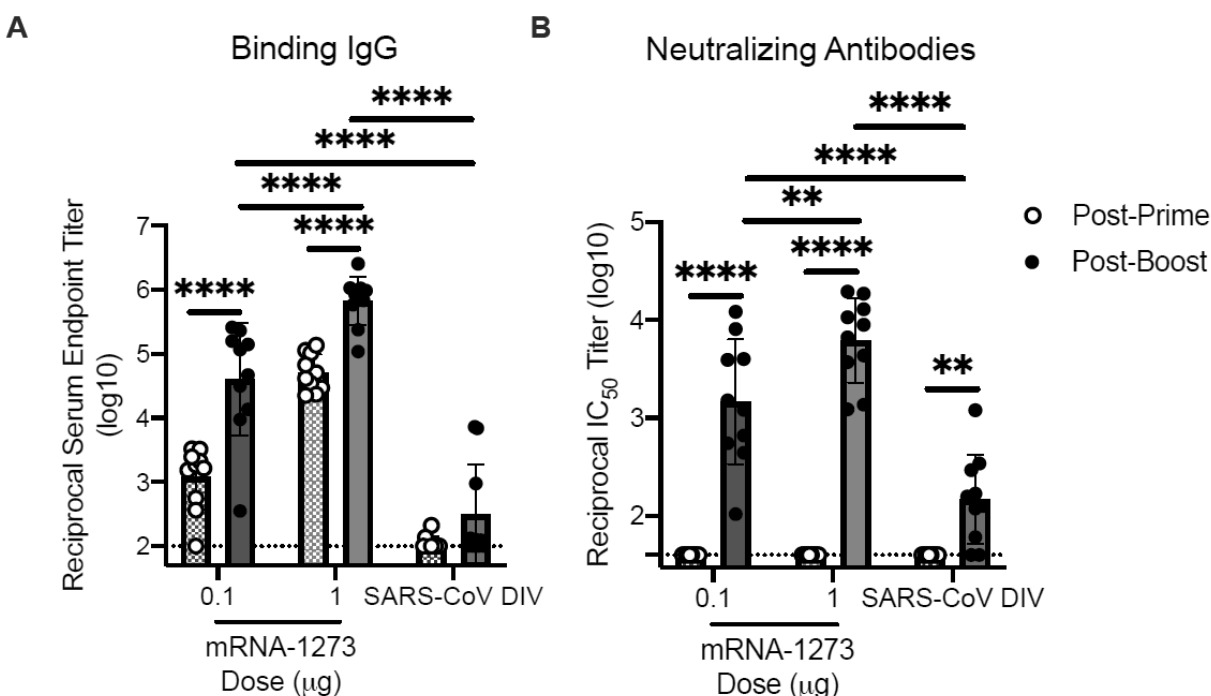
3. RESULTS

3.1. S-2P Specific IgG Antibody Responses and Neutralizing Activity

mRNA-1273 elicited robust S-2P-specific binding IgG antibody titers that increased in a dose-dependent manner after a prime/boost immunization schedule in aged mice (Figure 1A). Post-boost IgG antibody titers were significantly higher in mice immunized with 0.1 and 1 μ g of mRNA-1273 than in the SARS-CoV DIV group. Post-boost binding IgG antibody titers were also significantly higher than post-prime antibody titers in both the 0.1 and 1 μ g mRNA-1273 groups.

mRNA-1273 elicited a robust neutralizing antibody responses that increased in a dose-dependent manner after a prime/boost immunization schedule in aged mice (Figure 1B). Post-boost neutralizing antibody titers were significantly higher in the mRNA-1273 groups than in the SARS-CoV DIV group. A neutralizing antibody response was not detected post-prime in any of the treatment groups.

Figure 1: IgG Antibody Response and Neutralizing Antibody Response After mRNA-1273 and SARS-CoV DIV Treatment in Aged Mice



Abbreviations: IC₅₀ = concentration of vaccine producing 50% inhibition; IgG = immunoglobulin G; S-2P = spike protein with 2 proline substitutions within the heptad repeat 1 domain; SARS-CoV-2 = 2019 novel coronavirus; SARS-CoV DIV = double-inactivated severe acute respiratory syndrome coronavirus.

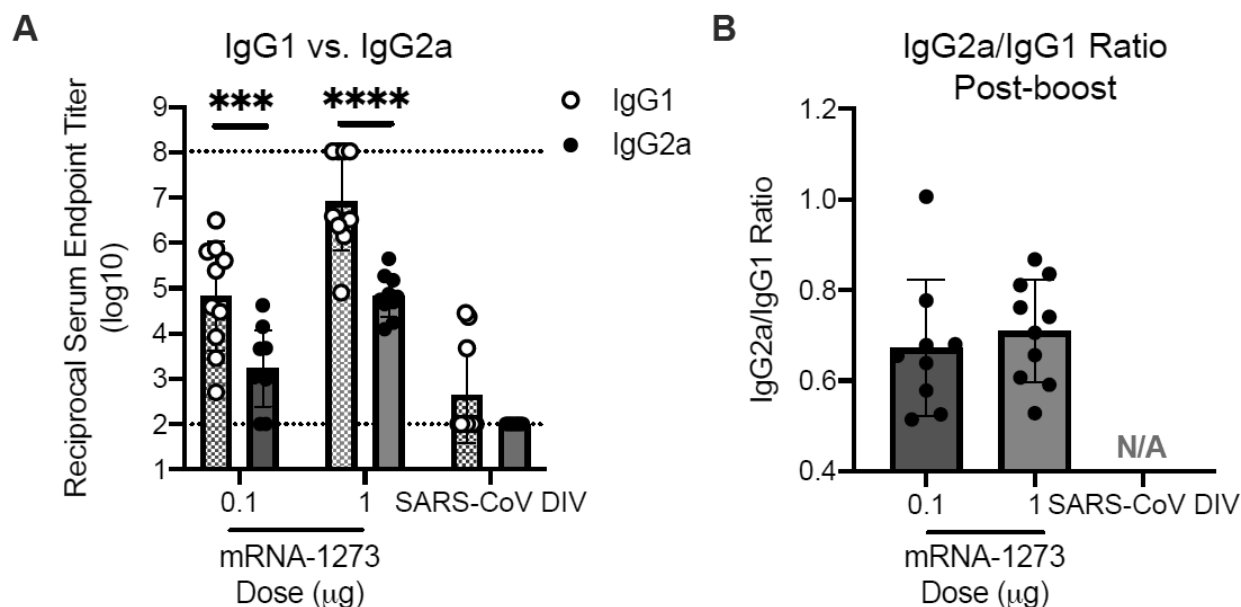
Note: A) Total IgG binding to S-2P; B) Neutralizing antibodies against SARS-CoV-2 pseudovirus. ** $p < .01$, **** $p < .0001$.

3.2. S-2P-Specific IgG Subclass Profile

In mice, a ratio of high IgG1 antibody subclass with low or no IgG2a antibody subclass is associated with a Th2-directed response, whereas a balanced ratio of IgG2a and IgG1 antibody subclasses is associated with a Th1-directed response. Post-boost, immunization with 0.1 and 1 μ g of mRNA-1273 elicited both IgG2a and IgG1 subclass S-2P-binding antibodies with a balanced IgG2a/IgG1 antibody subclass ratio. Notably, the IgG2a and IgG1 titers were higher in the 0.1 and 1 μ g mRNA-1273 groups than in the SARS-CoV DIV group (Figure 2). The IgG2a antibody subclass titers were not detectable in the SARS-CoV DIV group; therefore, the IgG2a/IgG1 ratio could not be calculated.

These results indicate that unlike SARS-CoV DIV, which induces a Th2-directed response that has been associated with ERD (Bolles et al 2011), mRNA-1273 is associated with a Th1-directed response that should not promote ERD.

Figure 2: IgG Antibody Response and Neutralizing Antibody Response After mRNA-1273 and SARS-CoV DIV Treatment in Aged Mice



Abbreviations: IgG = immunoglobulin G; SARS-CoV-2 = 2019 novel coronavirus; SARS-CoV DIV = double-inactivated severe acute respiratory syndrome coronavirus.

Note: A) IgG1 and IgG2a titers; B) Ratios of IgG2a to IgG1. *** $p < .001$, **** $p < .0001$.

3.3. Protection After SARS-CoV-2 MA10 Challenge

After challenge with SARS-CoV-2 MA10, weight loss was not observed in mice immunized with 1 μ g of mRNA-1273 through Day 4 post-challenge (Figure 3A). Mice vaccinated with PBS, SARS-CoV DIV, and 0.1 μ g of mRNA-1273 experienced weight loss by Day 4 post-challenge; however, the weight loss observed in the 0.1 μ g mRNA-1273 group was less than that observed in

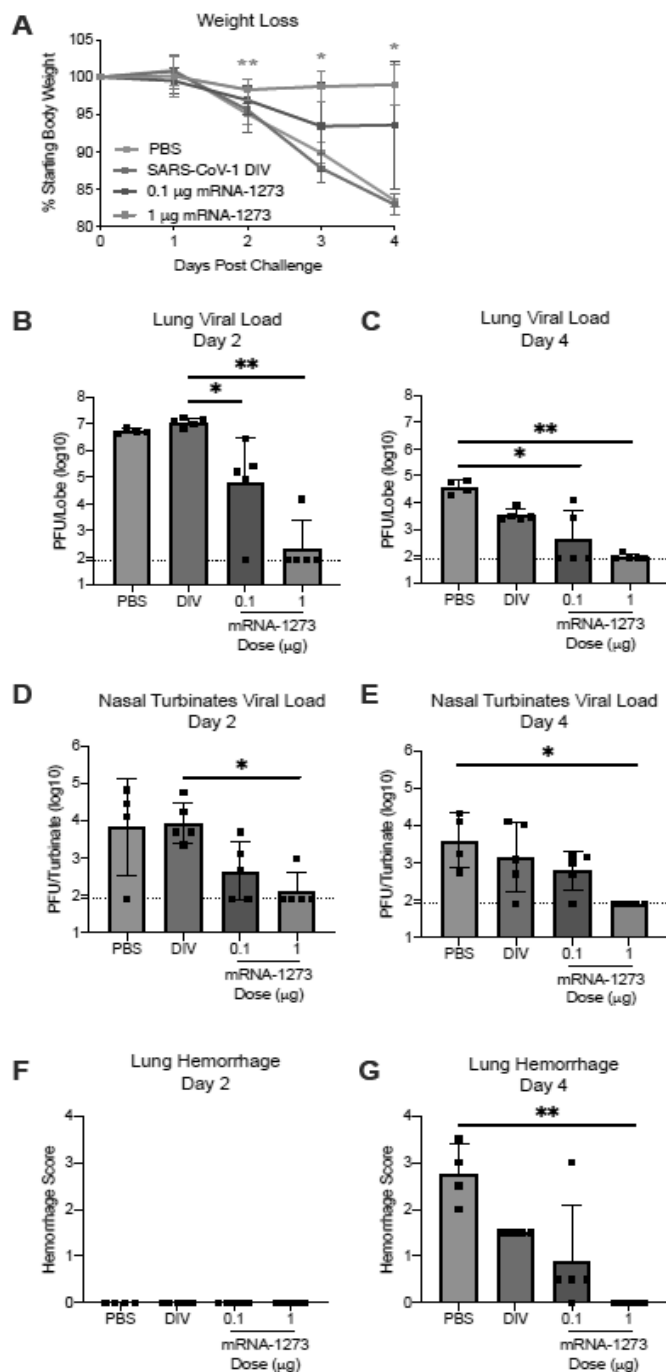
the SARS-CoV DIV and PBS groups, indicating that a suboptimal dose (0.1 µg) of mRNA-1273 did not enhance weight loss.

mRNA-1273 protected aged BALB/c mice from SARS-CoV-2 infection in a dose-dependent manner. Immunization with 1 µg of mRNA-1273 completely protected mice from viral replication on Day 2 and Day 4 post-challenge, with substantially lower viral titers in the lungs and nasal turbinates compared with all other groups. In fact, viral titers in the lungs and nasal turbinates were below the LOD at Day 4 post-challenge. Immunization with 0.1 µg of mRNA-1273 provided partial protection, as viral replication was observed in the lungs and nasal turbinates on Day 2 and Day 4 post-challenge, but the mean viral titers were lower than those in the SARS-CoV DIV and PBS groups. SARS-CoV DIV did not protect mice from SARS-CoV-2 infection, as viral replication was observed in the lungs and nasal turbinates and mean viral titers were similar to those in the PBS group on Day 2 and slightly lower than those in the PBS group on Day 4 post-challenge ([Figure 3B-E](#)).

No lung hemorrhage was observed in aged mice immunized with 1 µg of mRNA-1273 at Day 2 or Day 4 post-challenge. Lung hemorrhage was observed in mice in the 0.1 µg mRNA-1273, SARS-CoV DIV, and PBS groups on Day 4 post-challenge; however, the mean lung hemorrhage scores were lower in mice immunized with 0.1 µg of mRNA-1273 than in mice in the SARS-CoV DIV and PBS groups, indicating that a suboptimal dose (0.1 µg) of mRNA-1273 did not lead to lung hemorrhage enhancement ([Figure 3F](#) and [Figure 3G](#)).

Overall, an optimal dose (1 µg) of mRNA-1273 provided complete protection, a suboptimal dose (0.1 µg) of mRNA-1273 provided partial protection, and SARS-CoV DIV did not provide protection from SARS-CoV-2 infection in aged mice. Subprotective responses elicited by mRNA-1273 did not enhance weight loss, viral load, or lung hemorrhage.

Figure 3: Weight Change, Viral Load in Nasal Turbinates and Lungs, and Lung Hemorrhage Post-challenge in Aged Mice



Abbreviations: DIV/SARS-CoV-1 DIV = double-inactivated severe acute respiratory syndrome coronavirus;

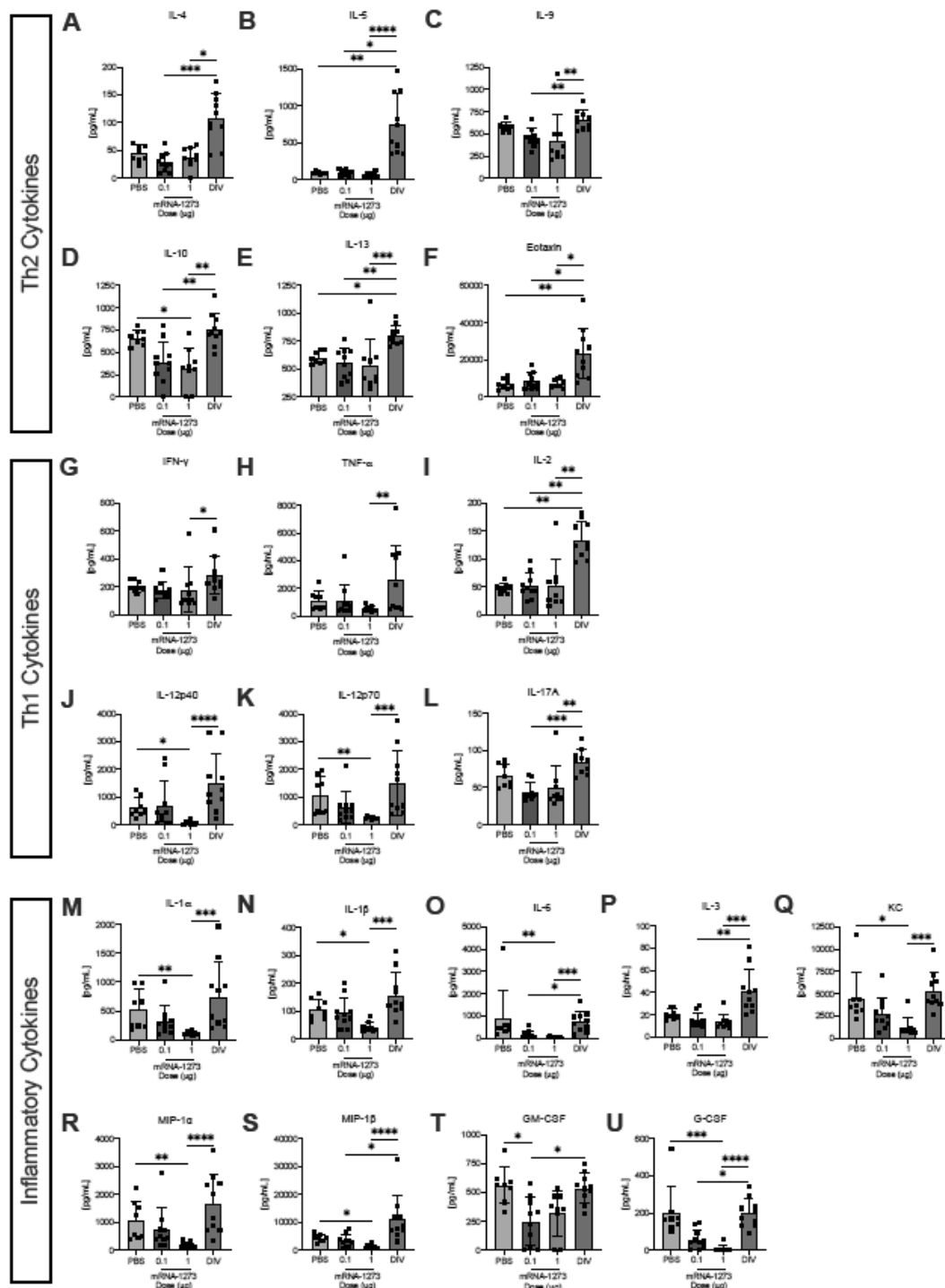
PBS = phosphate-buffered saline; PFU = plaque-forming unit.

Note: A) Summary of weight change through Day 4 post-challenge; B) Lung viral load at Day 2 post-challenge; C) Lung viral load at Day 4 post-challenge; D) Nasal turbinates viral load at Day 2 post-challenge; E) Nasal turbinates viral load at Day 4 post-challenge; F) Lung hemorrhage at Day 2 post-challenge; G) Lung hemorrhage at Day 4 post-challenge. * $p < .05$, ** $p < .01$. Lung hemorrhage was measured on a score ranging from 0 (no hemorrhage in any lobe) to 4 (extreme and complete hemorrhage in all lung lobes).

3.4. Lung Cytokines and Chemokines

Cytokines regulate the immune response to infection, and extensive changes in cytokine profiles are associated with the severity of disease.

The RT-PCR analysis showed that SARS-CoV DIV induced increased levels of lung cytokines and chemokines (Th2-associated cytokines and chemokines, Th1-associated cytokines, and inflammatory cytokines), and the expression levels of lung cytokines and chemokines were substantially higher at Day 4 post-challenge in the SARS-CoV group than the levels in mice immunized with mRNA-1273 ([Figure 4](#)). The expression levels of these cytokines in the SARS-CoV DIV group were generally higher than in the PBS group, whereas the expression levels in the 0.1 and 1 µg mRNA-1273 groups were generally lower than those in the PBS group. These results indicate that SAR-CoV DIV stimulates an enhanced Th2-associated cytokine response with the potential for respiratory disease enhancement, whereas mRNA-1273 induces a Th1-directed noninflammatory immune response suggestive of reduced respiratory disease.

Figure 4: Lung Cytokine and Chemokine Response After mRNA-1273 and SARS-CoV DIV Treatment in Aged Mice

Abbreviations: DIV = double-inactivated severe acute respiratory syndrome coronavirus; G-CSF = granulocyte colony-stimulating factor; GM-CSF = granulocyte-macrophage colony-stimulating factor; IFN = interferon; IL = interleukin; KC = keratinocyte chemoattractant; MIP = macrophage inflammatory protein; Th1 = T-helper 1; Th2 = T-helper 2; TNF = tumor necrosis factor.

Note: Analysis of: Th-2 Cytokines (A-F), Th-1 Cytokines (G-L) and Inflammatory cytokines (M-U). $p < .05$, ** $p < .01$, *** $p < .001$, **** $p < .0001$.

3.5. Lung Histopathology

Lung histology and immunohistochemistry results are presented in [Figure 5](#) and [Figure 6](#).

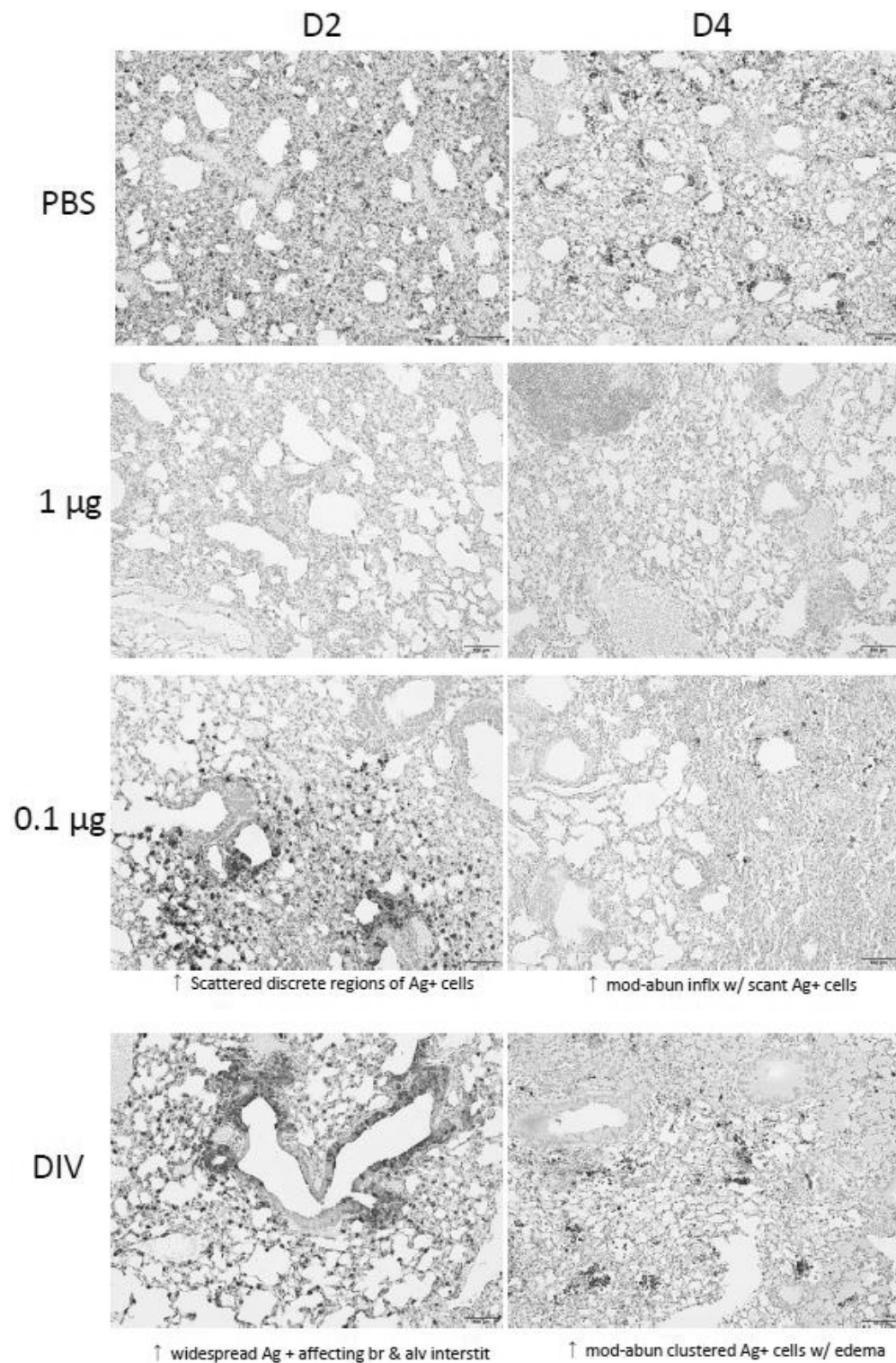
Phosphate-buffered saline group: Lung samples on Day 2 and Day 4 post-challenge showed similar pathology, characterized by a moderate to severe diffuse expansion of the alveolar capillaries and infiltration by large numbers of mononuclear and polymorphonuclear immune cells. Alveolar air spaces were substantially diminished, and moderate and diffuse Type II pneumocyte hyperplasia was observed. On Day 2 post-challenge, immunohistochemical evaluation for SARS-CoV-2 virus antigen showed abundant virus antigen-positive cells diffusely distributed throughout the alveolar interstitium. On Day 4 post-challenge, virus antigen-positive cells appeared concentrated and clustered and were intimately associated with regions surrounding the alveolar spaces.

mRNA-1273 1 µg group: Lung samples on Day 2 and Day 4 post-challenge showed similar pathology, characterized by a moderate diffuse expansion of the alveolar capillaries and infiltration by moderate numbers of mononuclear and polymorphonuclear immune cells. Alveolar air spaces were variably diminished, and moderate and diffuse Type II pneumocyte hyperplasia was observed. Immunohistochemical evaluation for SARS-CoV-2 virus antigen showed no virus antigen-positive cells.

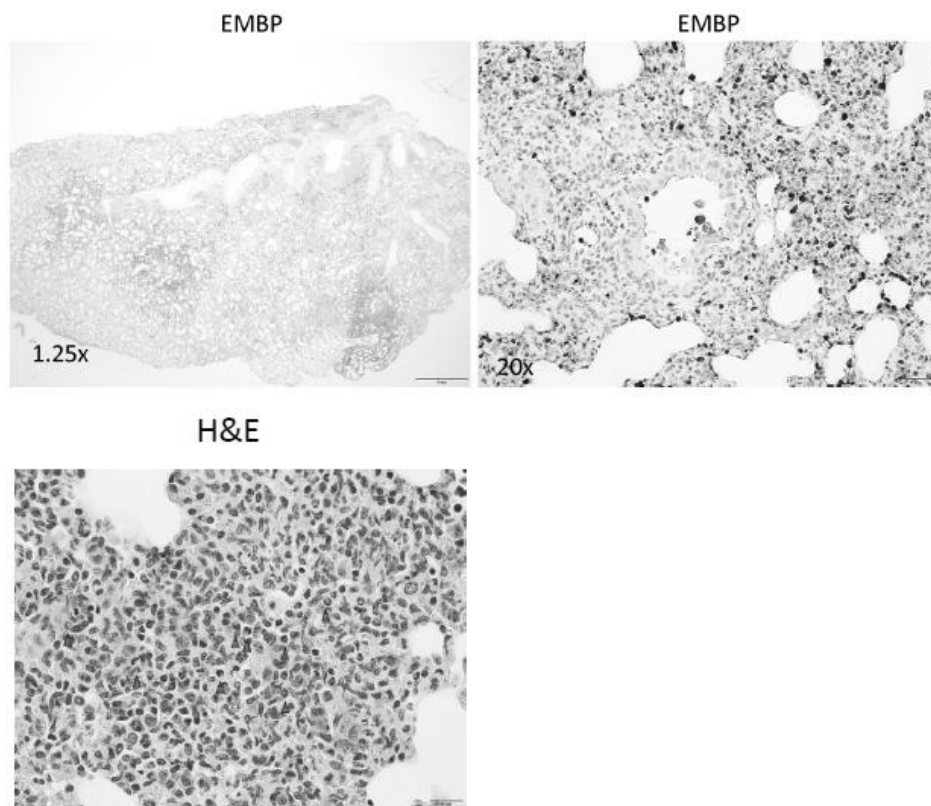
mRNA-1273 0.1 µg group: Lung sections on Day 2 post-challenge were characterized by multifocal and coalescing (occasionally discrete) regions of inflammation that closely circumscribed some small airways (bronchioles) and extended into the immediately adjacent interstitial areas. In these regions, abundant SARS-CoV-2 virus antigen-positive cells were concentrated; the antigen-positive cells included Type II pneumocytes, alveolar macrophages, and bronchiolar epithelium. On Day 4 post-challenge, virus antigen-positive cells were substantially decreased and were loosely scattered throughout hypercellular inflammatory foci.

SARS-CoV DIV group: Lung sections on Day 2 post-challenge were characterized by multifocal areas of mixed mononuclear and polymorphonuclear inflammatory cells that closely circumscribed medium and small airways (bronchioles) and extended into the immediately adjacent interstitial areas. In these regions, moderately abundant SARS-CoV-2 virus antigen-positive cells were concentrated; the antigen-positive cells included bronchiolar epithelial cells, Type II pneumocytes, and scattered alveolar macrophages. On Day 4 post-challenge, virus antigen-positive cells were densely concentrated and clustered/clumped throughout the pulmonary interstitium. These clustered antigen-positive immune cells were closely associated with and occasionally lined the alveolar spaces. Substantial accumulations of fluid (edema) and fibrin were observed within alveolar spaces, and Type II pneumocyte hyperplasia was also present.

Figure 5: Lung Histopathology at Day 2 and Day 4 Post-challenge in Mice Administered PBS, 0.1 or 1 µg of mRNA-1273, or SARS-CoV DIV



Abbreviations: abun = abundant; Ag = antigen; alv = alveoli; br = bronchioles; D = day; DIV = double-inactivated severe acute respiratory syndrome coronavirus; interstit = interstitial; mod = moderate; PBS = phosphate-buffered saline; SARS-CoV DIV = double-inactivated severe acute respiratory syndrome coronavirus.

Figure 6: Lung Histopathology on Day 2 Post-challenge: SARS-CoV DIV Group

Abbreviations: EMBP = eosinophil major basic protein; H&E = hematoxylin and eosin; SARS-CoV DIV = double-inactivated severe acute respiratory syndrome coronavirus.

4. CONCLUSIONS

mRNA-1273 is highly immunogenic in aged (approximately 1-year-old) mice. mRNA-1273 elicited robust binding and neutralizing antibody responses in a dose-dependent manner after a prime/boost immunization schedule. The mRNA-1273 groups showed significantly higher SARS-CoV-2 S-2P IgG antibody titers and neutralizing antibody responses post-boost compared with the SARS-CoV DIV group.

Immunization with 0.1 and 1 µg of mRNA-1273 elicited both IgG2a and IgG1 subclass S-2P-binding antibodies post-boost, with a balanced IgG2a/IgG1 response. In contrast, SARS-CoV DIV elicited IgG1 subclass S-2P-binding antibodies post-boost, but IgG2a subclass S-2P-binding antibodies were not detected and the corresponding IgG2a/IgG1 ratio could not be calculated. These results indicate that unlike SARS-CoV DIV, which induces a Th2-directed response that has been associated with ERD ([Bolles et al 2011](#)), mRNA-1273 is associated with a Th1-directed response that does not promote ERD.

mRNA-1273 protected aged BALB/c mice from SARS-CoV-2 infection in a dose-dependent manner. Immunization with 1 µg of mRNA-1273 completely protected the mice after challenge with SARS-CoV-2 MA-10. Weight loss was not observed through Day 4 post-challenge, viral titers in the lungs and nasal turbinates were below the LOD at Day 4 post-challenge, and no lung hemorrhage was observed at Day 2 or Day 4 post-challenge. Immunization with 0.1 µg of mRNA-1273 provided partial protection, characterized by weight loss, viral replication, and lung hemorrhage that was less than what was observed in the SARS-CoV DIV and PBS groups. These results indicate that subprotective responses elicited by immunization with a suboptimal dose of mRNA-1273 (0.1 µg) did not enhance weight loss, reduced viral load and did not lead to lung hemorrhage enhancement.

Cytokine levels in the lungs are indicative of viral disease pathogenesis and severity of the infection. Both poor outcome and poor immune response likely correlate with impaired cytokine responses with an unbalanced Th1- and Th2-associated cytokine response. Th2-associated cytokine and chemokine expression and inflammatory cytokine expression in the SARS-CoV DIV group were higher than those in the PBS group, indicating an exacerbated cytokine response suggestive of ERD. In contrast, mRNA-1273 induced a balanced Th1-directed cytokine response with lower expression of inflammatory cytokines than in the SARS-CoV DIV and PBS groups, indicating a noninflammatory immune response.

Lung histopathology and immunochemistry showed no alterations related to ERD or inflammation and no virus antigen-positive cells in mice immunized with 1 µg of mRNA-1273 on Day 2 or Day 4 post-challenge. Lung sections in mice immunized with 0.1 µg of mRNA-1273 showed inflammation and SARS-CoV-2 virus antigen-positive cells on Day 2 post-challenge; however,

virus antigen-positive cells were substantially reduced on Day 4 post-challenge. The SARS-CoV DIV and PBS groups showed marked alterations related to enhanced disease and inflammation with abundant SARS-CoV-2 virus antigen-positive cells.

Overall, mRNA-1273 is immunogenic and efficacious and does not promote ERD in aged mice.

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6. REPORT APPROVAL

Study Director:

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Date: 2020.10.28 17:23:05 -04'00'

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Viral Pathogenesis Laboratory, VRC, NIH
Building 40, Room 2608
Bethesda, MD 20892.

Date

7. APPENDICES

Appendix 1 [Summary of Analysis: Lot AMPDP-200005 Release](#)



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phone 617-714-6500 • fax 617-583-1998

SUMMARY OF ANALYSIS

Product Description: mRNA-1273 Drug Product

Project Code: mRNA-1273 **Batch ID:** B-20200201-01404

Manufacturer: Moderna **Sample ID:** S-200201-00010

Moderna Lot Number: AMPDP-200005 **Date of Manufacture:** 01-Feb-2020

mRNA Lot Number: (b) (4) **Expiry Date:** N/A

Volume: 0.6 mL **Shipping Temperature:** -70°C

Storage Temperature: -70°C

Final Buffer 10.7 mM Sodium Acetate; 8.7% Sucrose; 20 mM Tris, pH 7.5

Container Closure: 2R vial, USP Type I; Traditional rubber stoppers are West 4432/50 (chlorobutyl rubber) composition with a Flurotec coating.

Test	Method	Target Attributes	Results
Identity	DPTM-0023	(b) (4)	(b) (4)
Appearance	DPTM-0030	White to off-white dispersion. May contain visible, white or translucent product-related particulates	Conforms, White to off-white dispersion; essentially free of visible particulates
RNA Content	DPTM-0024	(b) (4)	(b) (4)
RNA Purity	DPTM-0046	(b) (4)	(b) (4)
% Encapsulation	DPTM-0063	(b) (4)	(b) (4)
Filtered Mean Particle Size	DPTM-0012	(b) (4)	(b) (4)
Filtered Polydispersity Index		(b) (4)	(b) (4)



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Test	Method	Target Attributes	Results		
Lipid Identification:					
SM-102	DPTM-0019	(b) (4)	(b) (4)		
Cholesterol					
DSPC					
PEG2000-DMG					
Lipid Content (mg/mL)					
SM-102	DPTM-0019	(b) (4)		(b) (4)	
Cholesterol					
DSPC					
PEG2000-DMG					
Lipid Impurities (% Area)					
Individual Impurities	DPTM-0019	(b) (4)			(b) (4)
Total Impurities					
pH	USP <791> DPTM-0028				
Osmolality	USP <785> DPTM-0029				
Bacterial Endotoxin	USP <85>				
Bioburden	USP<61>				
Particulate Matter	USP <788> Method 2				



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REVISION HISTORY

Revision #	Change Details	Author	Effective Date
1.0	Introduction of a New Document	(b) (6)	Date of Approval
2.0	Added in results for Identity and particulate matter	(b) (6)	Date of Approval
3.0	Added in results for bioburden testing and updated bacterial endotoxin results with values from ACCI (DPAD-00577)	(b) (6)	Date of Approval
4.0	Added correct test methods for RPIP and (b) (4)	(b) (6)	Date of Approval

Appendix 2 Immunogenicity and Efficacy of mRNA-1273 and SARS-CoV Double-Inactivated Virus (DIV) in Aged BALB/c Mice

**Immunogenicity and Efficacy of mRNA-1273 and SARS-CoV Double-Inactivated Virus (DIV)
in Aged BALB/c Mice**

(b) (6)

Lead, Coronavirus Vaccines and Immunopathogenesis Team
NIH, NIAID, Vaccine Research Center, Viral Pathogenesis Laboratory
June 29, 2020

Principal Investigator: (b) (6)

Program Managers: (b) (6)

ELISAs: (b) (6)

Pseudovirus Neutralization Assay: (b) (6)

Data Analyses: (b) (6)

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Immunization, challenge, and lung cytokine analyses completed by (b) (4)

(b) (4)

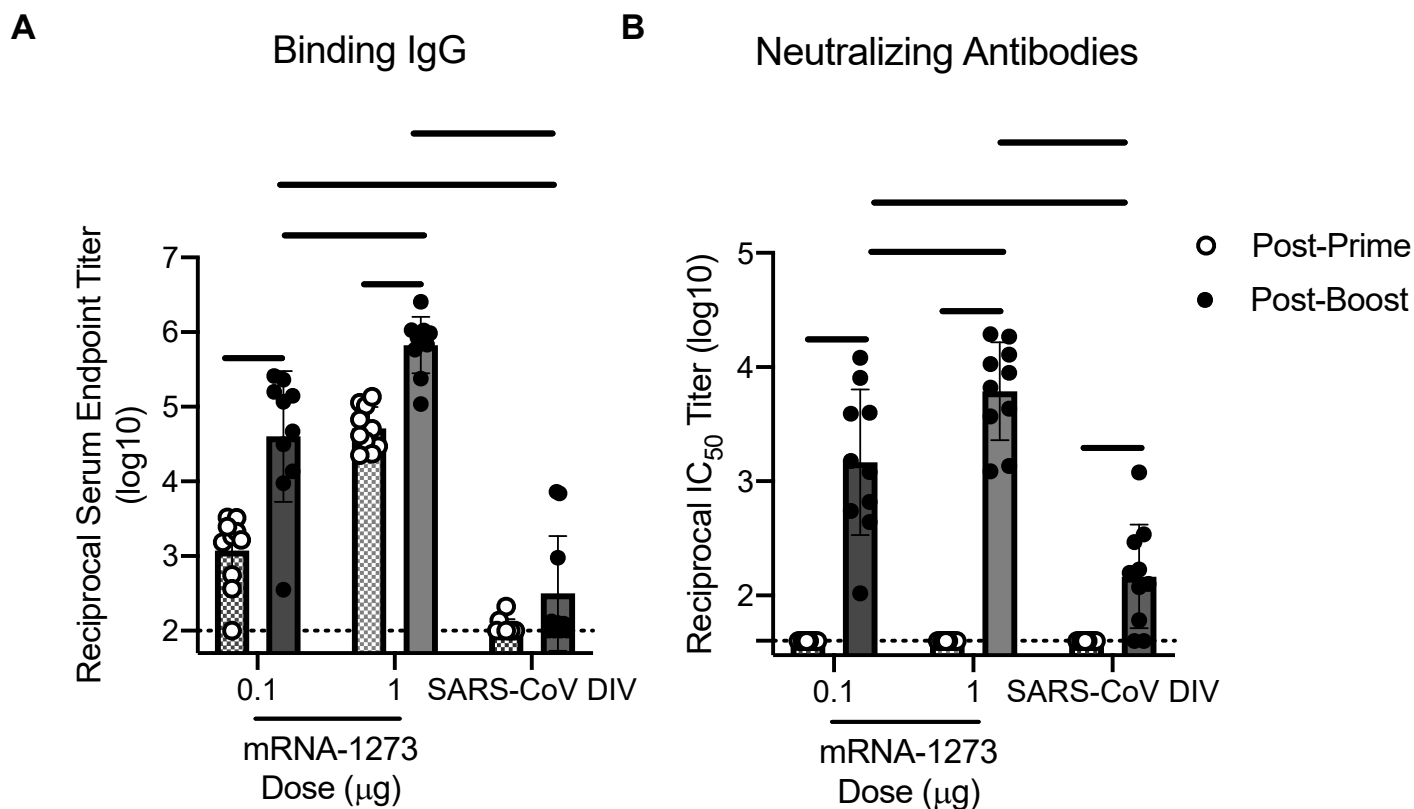


Figure 1. mRNA-1273 elicits robust antibody responses in aged BALB/c mice. Aged (> 1 year old) BALB/c mice were immunized, at weeks 0 and 3, with 0.1 (blue) or 1 µg (red) of mRNA-1273. Mice were also immunized with a historical dose (10 µL of stock) of SARS-CoV double-inactivated virus (DIV) (purple) to assess Th2-skewed immunity. Sera were collected 2 weeks post-prime (unfilled circles) and post-boost (filled circles) and assessed for SARS-CoV-2 S-specific total IgG by ELISA (A) and neutralizing antibodies against SARS-CoV-2 pseudovirus (B). Each symbol represents an individual mouse. Bars represent GMT, and error bars indicate geometric standard deviation (SD). Dotted line indicates assay limit of detection. Two-way ANOVA with Sidak's or Tukey's multiple comparisons test was used to compare temporal and dose-dependent antibody responses post-boost, respectively. ** = p-value < 0.01, **** = p-value < 0.0001.

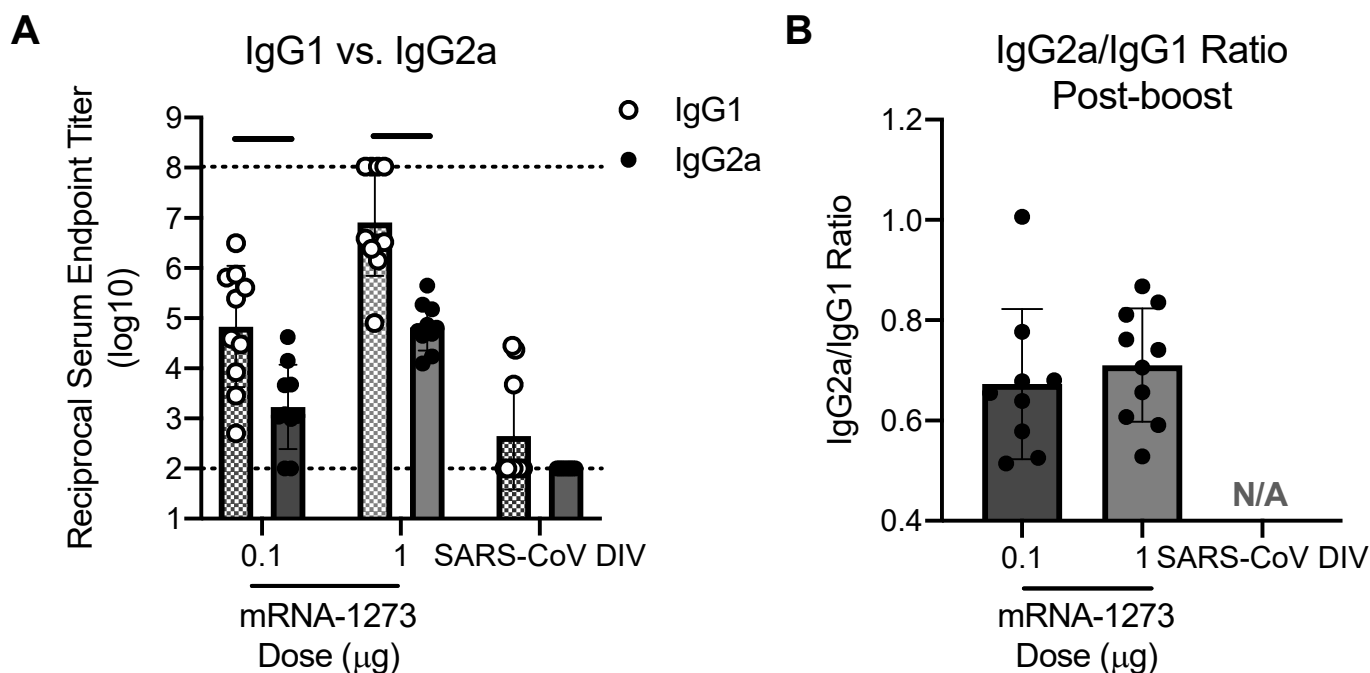


Figure 2. mRNA-1273 elicits both IgG2a and IgG1 subclass spike-binding antibodies in aged BALB/c mice. Aged BALB/c mice were immunized, at weeks 0 and 3, with 0.1 (blue) or 1 µg (red) of mRNA-1273. Mice were also immunized with a historical dose (10 µL of stock) of SARS-CoV DIV (purple) to assess Th2-skewed immunity. Sera were collected 2 weeks post-boost and assessed by ELISA for SARS-CoV-2 S-specific IgG1 and IgG2a. Endpoint titers (A) and endpoint titer ratios of IgG2a to IgG1 (B) were calculated. For endpoint titers that did not reach the lower limit of detection (dotted line), ratios were not calculated. Each symbol represents an individual mouse. (A) Bars represent geometric mean titers (GMT), and error bars indicate geometric SD. Two-way ANOVA with Sidak's multiple comparisons test was used to compare IgG subclasses. (B) Bars represent means, and error bars indicate arithmetic SD. No significance between mRNA-1273 doses was found by T-test. *** = p-value < 0.001, **** = p-value < 0.0001.

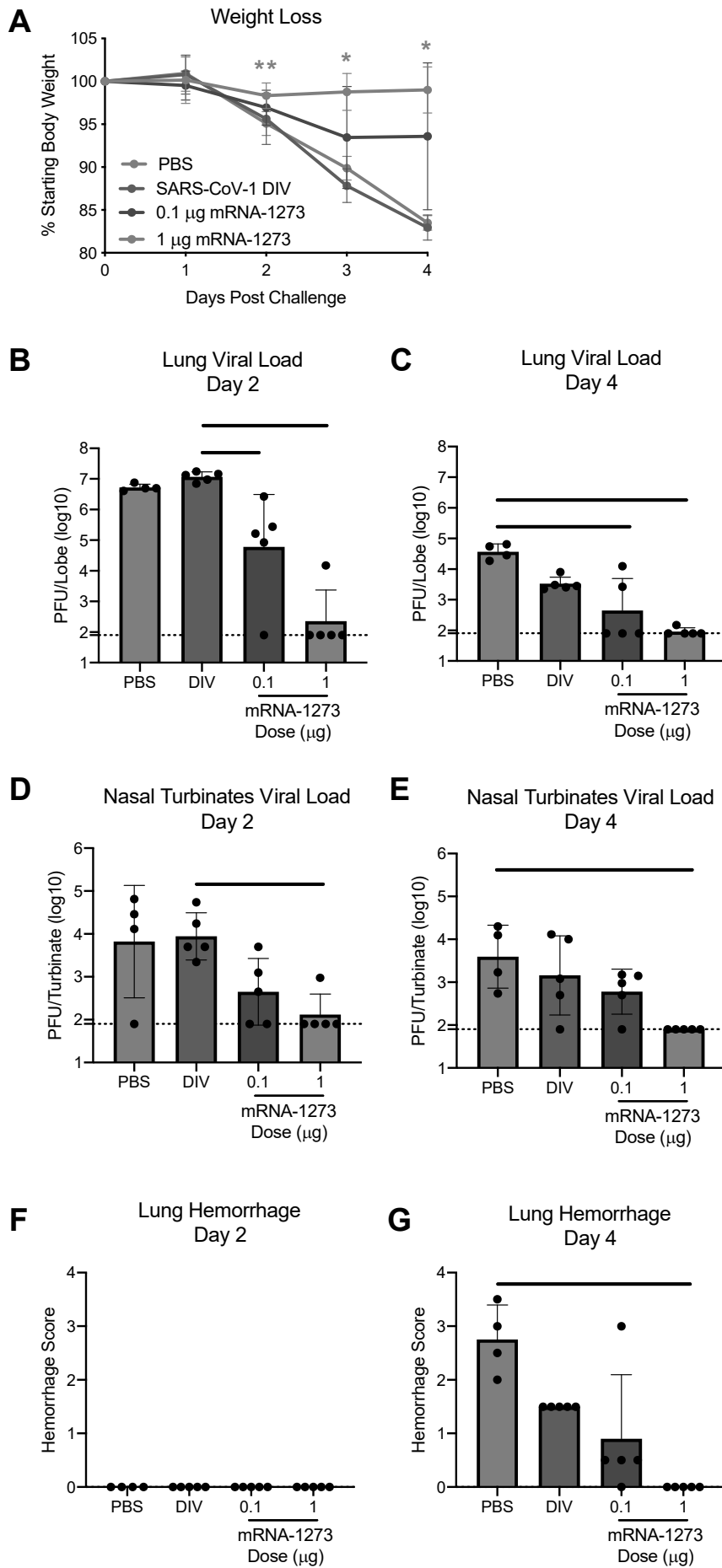


Figure 3. mRNA-1273, but not SARS-CoV DIV, protects mice from disease following SARS-CoV-2 challenge. Aged BALB/c mice were immunized at weeks 0 and 3, with 0.1 (blue) or 1 µg (red) of mRNA-1273. Mice were also immunized with a historical dose (10 uL of stock) of SARS-CoV DIV (purple) to assess Th2-skewed immunity. Mock-immunized mice were immunized with PBS (gray). Four weeks post-boost, mice were challenged with 1×10^3 PFU of mouse-adapted SARS-CoV-2 that has been optimized for increased disease manifestation in mice by 10 *in vivo* passages (SARS-CoV-2 MA-10). Following challenge, mice were monitored for weight loss (A). At days 2 and 4 post-challenge, lungs (A-B) and nasal turbinates (C-D) from 5 mice/group were harvested for analysis of viral titers. Lung hemorrhage (0 = no hemorrhage, 4 = severe hemorrhage in all lobes) was also assessed (F-G). (A) The mean of each group is represented by circles. Error bars represent arithmetic SDs. Weight loss in experimental groups was compared to the PBS-immunized group on each day post-challenge by one-way ANOVA with Kruskal-Wallis post-test. (B-G) Each dot represents an individual mouse; bars represent the GMT (B-E) or mean (G-F) of each group, and error bars represent geometric or arithmetic SD, respectively. Groups were compared by one-way ANOVA with Kruskal-Wallis post-test. * = p-value < 0.05, ** = p-value < 0.01.

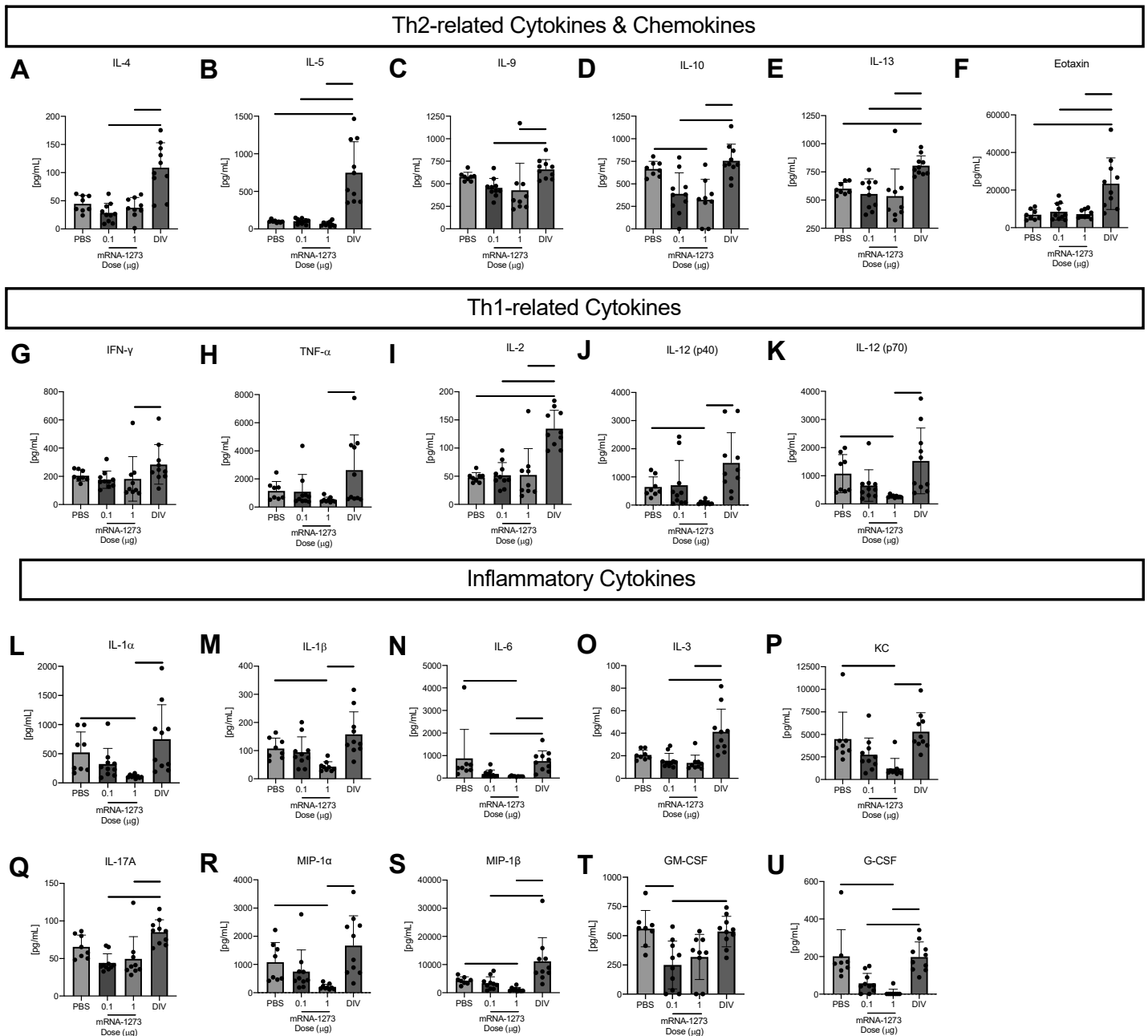


Figure 4. DIV induced increased levels of lung cytokines compared to mRNA-1273 in SARS-CoV-2 challenged aged BALB/c mice. BALB/c mice were immunized, at weeks 0 and 3, with 0.1 (blue) or 1 μ g (red) of mRNA-1273. Mice were also immunized with a historical dose (10 μ L of stock) of SARS-CoV DIV (purple) to assess Th2-skewed immunity. Control mice were immunized with PBS (gray). Four weeks post-boost, mice were challenged with 1×10^3 PFU of SARS-CoV-2 MA-10. Four days following challenge, lung supernatants were harvested (N = 5 mice/group) for analysis of lung cytokines by (b) (4). Cytokine concentrations were interpolated from standard curves for each cytokine. Dots represent individual mice, bars represent the mean of each group, and error bars represent arithmetic SD. Groups were compared by one-way ANOVA with Kruskal-Wallis post-test. * = p-value < 0.05, ** = p-value < 0.01, *** = p-value < 0.001, **** = p-value < 0.0001.

Results and Interpretation:

- mRNA-1273 elicits robust binding and neutralizing antibodies in aged BALB/c mice, with balanced IgG2a/IgG1 responses, as seen in young adult BALB/c mice.
- mRNA-1273 protects aged BALB/c mice from SARS-CoV-2 disease in a dose-dependent manner, with breakthrough at the 0.1 µg dose.
- Subprotective responses elicited by mRNA-1273 did not lead to enhanced weight loss, viral load, or lung hemorrhage.
- SARS-CoV DIV does not protect aged BALB/c mice from SARS-CoV-2 disease and does not lead to enhanced weight loss, viral load, or lung hemorrhage.
- SARS-CoV DIV induced increased levels of lung cytokines in SARS-CoV-2-challenged aged BALB/c mice, whereas mRNA-1273 did not; this data suggests mRNA-1273 induces Th1-skewed non-inflammatory immune responses.

Future Direction:

- Lungs will be examined by a veterinary pathologist for virus and immune cell infiltrates to assess VAERD signals.